How to Read Empirical Articles

Reading empirical articles for the first time can be daunting. Technical jargon, statistical analyses, and discussions of theories and findings you know little about (at least initially) all contribute to feelings of paralysis when you begin reading an article. However, it doesn't have to be this way! The following tips can help to get you through an empirical article.

Before reading an article ask yourself Why am I reading this and what do I want to get out of it? By first identifying *why* you are reading the article you can better target *how* you read the article.

Remember that you are not reading a novel where you have to start at the first word, end at the last and read every word in between. Based on your needs, focus on the relevant sections.

In this course, you are reading articles to evaluate whether a study's findings actually support the authors' conclusions. Focus on understanding what the researchers did, what they found, and whether the conclusions follow from the results. This is the most important skill to develop as a critical reader.

In future courses, you might read empirical articles for other reasons:

- Are you reading it for a course where the instructor wants you to get something specific out of it (e.g., general knowledge of the field, a critical evaluation of an idea)? How much other reading has been assigned and how much time can you devote to each paper? Don't spend 3 hours reading the first 2 papers and 10 minutes reading the last 8. Don't assume that short papers are easier to read than long ones.
- Are you looking for research ideas? Read the introduction to see what's been done, the discussion to see what questions remain, and read the methods critically to look for flaws in the logic or design.
- Are you reading the article to plan your own experiment? Pay particular attention to the methods section. Take notes.

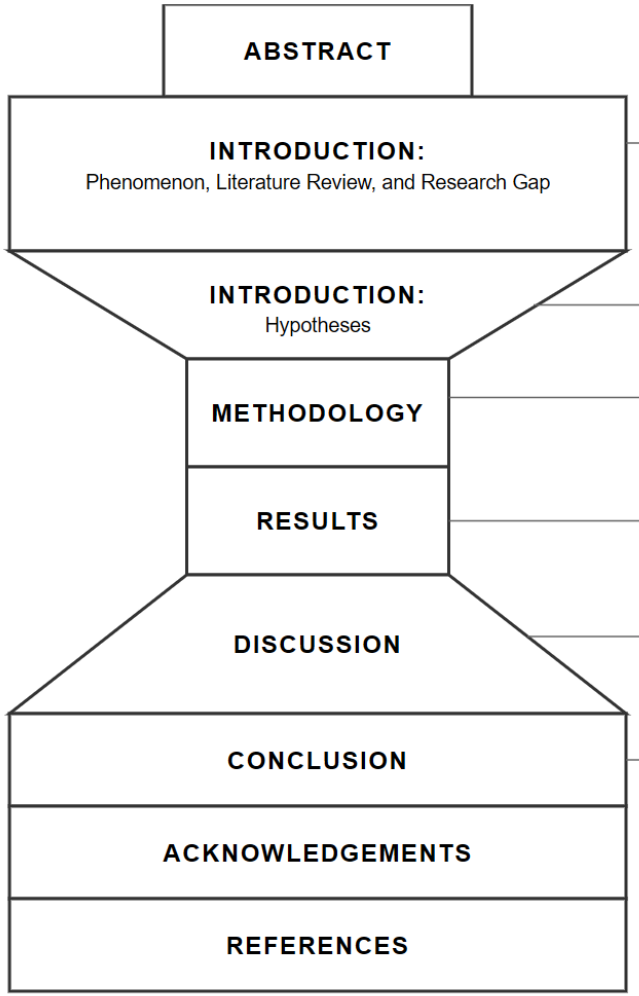
Empirical Article Sections

Each empirical article has a standard set of sections. Understanding their purpose will help you know what to expect and where to find specific information. The major sections are: abstract, introduction, methods, results, and discussion. The American Psychological Association (APA) Publication Manual, which sets formatting standards for psychology papers, describes these sections in detail. Here are the basics:

Section	Function
Abstract	A single paragraph summarizing the entire article. Includes hypotheses, methods, results & implications. Usually appears at the top of the article in a separate block before the main text.
Introduction	Introduces phenomenon to be studied, provides background research, explains hypotheses.
Methods	Describes how the study was actually conducted, including participants, measures used, and procedure.
Results	Describes what was found in the study and whether the hypotheses were disconfirmed or confirmed. Should include both numbers (statistics) and verbal explanations.
Discussion	Describes the implications of the study, addresses study limitations, suggests future research.

The diagram on the next page shows how these sections fit together. Articles start broad, narrow to the specific study, then broaden again to discuss what it means.

Typical Research Article Structure



Questions to ask while reading:

What are they interested in studying?
Does the literature review support the claim that something is missing or unclear?
Is the research gap significant?

What are the hypotheses based on?

How were the key concepts measured (operationalized)?
Are the elements of the experiment (subjects, stimuli, etc.) representative of the phenomenon being studied?

Do the results support or contradict the hypotheses?
(For now, you can assume the statistics are correct.)

Do the authors' claims go beyond what the data actually showed?
Are the authors careful about moving from operationalized concepts to ordinary ones?

Do the conclusions match the specific findings, or do they make broader claims than the evidence supports?

Tips for Reading an Empirical Article

1. Get the core ideas first. Read the abstract, and the first and last few paragraphs of the introduction and discussion section. This will give you a good idea of what to expect from the piece as a whole (phenomenon being studied; hypotheses; main results; implications). Be especially attentive to the hypotheses of the article. What are the authors trying to test? For example, imagine researchers are testing the effect of caffeine on test performance. The independent variable is the factor they manipulate (caffeine), and the dependent variable is the outcome they measure (test performance).

2. Read for detail. Once you feel you have a general sense of the article's direction, read the introduction and discussion more deeply. What led them to their hypothesis? What did they find? How does this contribute to our understanding of phenomenon *X*? Note any statements by the authors (or your own thoughts) about how this article changes or augments the original theory.

3. Put yourself in the participants' shoes. While reading the methods section, try to imagine, as vividly as possible, being a participant in the experiment. This will help to make more concrete the actual experiment. Understand how the conceptual independent and dependent variables were actually manipulated and measured in the study (i.e., operationalized).

4. Don't get bogged down in the statistics, but do understand what was found. If you feel comfortable with statistics, by all means puzzle through all the analyses in the results section. If you're less experienced with statistics, keep in mind that most good articles will first present the numbers, then explain what the results mean in regular language. Focus on extracting the meaning of the results. Also, look at any tables and graphs, and how they are described in the text.

5. At least one more read through. A little while after initially reading the article, go back and skim through everything again. See if it all fits together now that you've had time to reflect on the piece. Note anything new you noticed or realized about the article.

Adapted from

- Culham, J. (2006). *How to Read a Journal Article*. Retrieved March 16, 2026 from http://www.culhamlab.com/how-to-read-a-journal-article.
- Jordan, C. H. & Zanna, M. P. (1999). How to read a journal article in social psychology. In R. F. Baumeister (Ed.), *The self in social psychology* (pp. 461-469). Philadelphia, PA: Psychology Press.
- Oyamot, C. *Reading Empirical Articles: Some Tips*. Retrieved March 16, 2026 from https://web.archive.org/web/20081012175405/http://www.psych.sjsu.edu/~neldred/Dr.%20Oyamot's%20How%20To%20Read%20An%20Empirical%20Article.htm.